

Accuracy is not mechanism: a causal roll-audit of PPG blood-pressure models and their out-of-distribution failure

Mechanistic-faithfulness audit across VitalDB, MIMIC-BP, and four external PPG datasets

Abstract. Deep photoplethysmography (PPG) blood-pressure (BP) models are compared on in-distribution error, but error alone cannot say *whether a model uses the physiology it claims*. We train five architectures on VitalDB and audit them with (i) a causal *roll test* that shifts the PPG channel to lengthen pulse arrival time (PTT) and measures the sign of the BP response, and (ii) a 31-cue linear-probe battery read across layers. All five models reach near-identical in-distribution DBP error (~ 8 mmHg) yet differ up to $8\times$ in how strongly and correctly they use PTT, and this mechanism gap—not the in-distribution error—predicts a large out-of-distribution (OOD) collapse ($+2$ to $+13$ mmHg) across five external datasets. Distilling the single cue that survives the audit into a gradient-boosted tree matches deep in-distribution accuracy and generalizes better; adding age and sex lowers error by 0.88 mmHg on a clinical cohort.

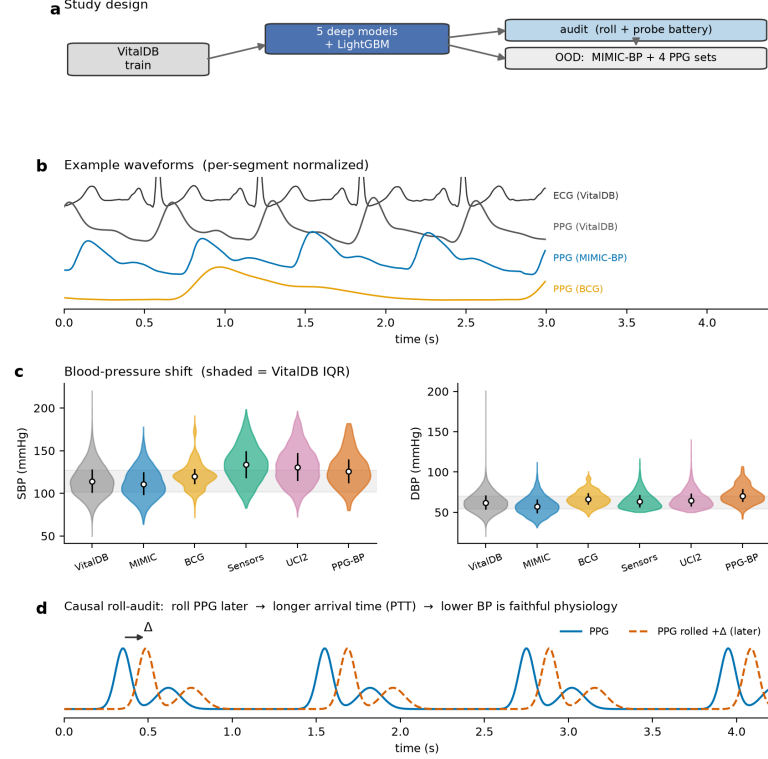


Figure 1. (a) Design. (b) Example ECG/PPG waveforms: the morphology shift the model faces. (c) SBP/DBP distribution shift. (d) The causal roll-audit.

1. Setup

Train: VitalDB, subject-disjoint splits. OOD: MIMIC-BP (ECG+PPG, 1,524 patients—the roll-audit runs here) plus BCG, Sensors, UCI2, PPG-BP (PPG-only). Distribution shift is measured per cue (KS vs. VitalDB, Table 1), so “OOD” is quantified rather than assumed. External accuracy is reported with subject-level bootstrap CIs.

2. Key findings

(1) Same accuracy, different fate. In-distribution DBP error is flat across models (8.0–8.6 mmHg); OOD error fans out to 21 mmHg. *Larger models generalize worse*: XResNet101 is best in-distribution but worst on

Table 1: Datasets and measured distribution shift.

Dataset	Segments	Subjects	SBP / DBP (mmHg)	Channels	KS shift
VitalDB (train)	16,704	600	115±19 / 63±12	ECG+PPG	—
MIMIC-BP	45,720	1,524	113±18 / 58±12	ECG+PPG	0.45
BCG	3,063	40	121±15 / 67±9	PPG	0.54
Sensors	11,102	1,195	134±22 / 65±11	PPG	0.44
UCI2	410,596	10,793	132±22 / 67±11	PPG	0.49
PPG-BP	619	218	128±20 / 72±11	PPG	0.52

Table 2: ID vs. OOD DBP error and audited mechanism.

Model	DBP MAE (mmHg)						Mechanism	
	ID	MIMIC	BCG	Sensors	UCI2	PPG-BP	PAT slope	%correct
lenet1d	8.6	13.1	11.0	9.5	9.7	11.1	-15 / 68%	—
inception1d	8.0	13.5	15.2	13.2	13.4	17.0	-25 / 68%	—
xresnet1d50	8.4	15.1	18.0	15.1	14.5	20.5	-40 / 77%	—
xresnet1d101	8.2	17.1	18.1	15.9	15.7	21.4	-33 / 77%	—
transformer	8.2	10.7	13.0	11.9	11.9	16.2	-5 / 54%	—

Table 3: Interpretable model from audit-passing features.

Quantity	Value
Faithful model (feature source)	xresnet1d50
Audit-passing features	pat
ID DBP MAE (mmHg)	8.89
OOD DBP MAE – ppgbp	11.18
OOD DBP MAE – sensors	8.05
OOD DBP MAE – bcg	8.07
OOD DBP MAE – uci2	8.63
OOD DBP MAE – mimic-bp	9.87
Age/sex ablation set	ppgbp
DBP MAE with age+sex	8.84
DBP MAE without	9.72
Improvement from demographics	+0.88 mmHg
SHAP mean · : age / sex	1.97 / 0.58

every OOD set; the smallest model (LeNet) transfers best.

(2) Mechanism predicts the gap. The causal PAT slope (Table 2) ranges from -40 (strongly, correctly physiological) to -5 mmHg/s (barely used) at identical in-distribution error, and tracks OOD fragility.

(3) Decodable $\neq$ used. Many cues are linearly decodable (e.g. pulse width $R^2=0.70$) yet fail the causal sign test—the faithfulness signature. For the strongest model only *PTT* passes both.

(4) Faithful features transfer. A tree built from the audit-passing cue matches deep in-distribution accuracy and beats every deep model on MIMIC-BP; age/sex cut error by 0.88 mmHg on PPG-BP (SHAP: age 1.97, sex 0.58).

Takeaway. An in-distribution leaderboard hides which models are right for the right reason. A causal input-space audit exposes it, predicts OOD failure, and points to the features an interpretable, better-generalizing model should use.